

RoboPainting

Final Presentation

(Built on indirect force control)

Group 13

Ruohan He

Shixuan Sun

Jiawen Wang

Xiaoyang Zhao



Overview



GOAL:

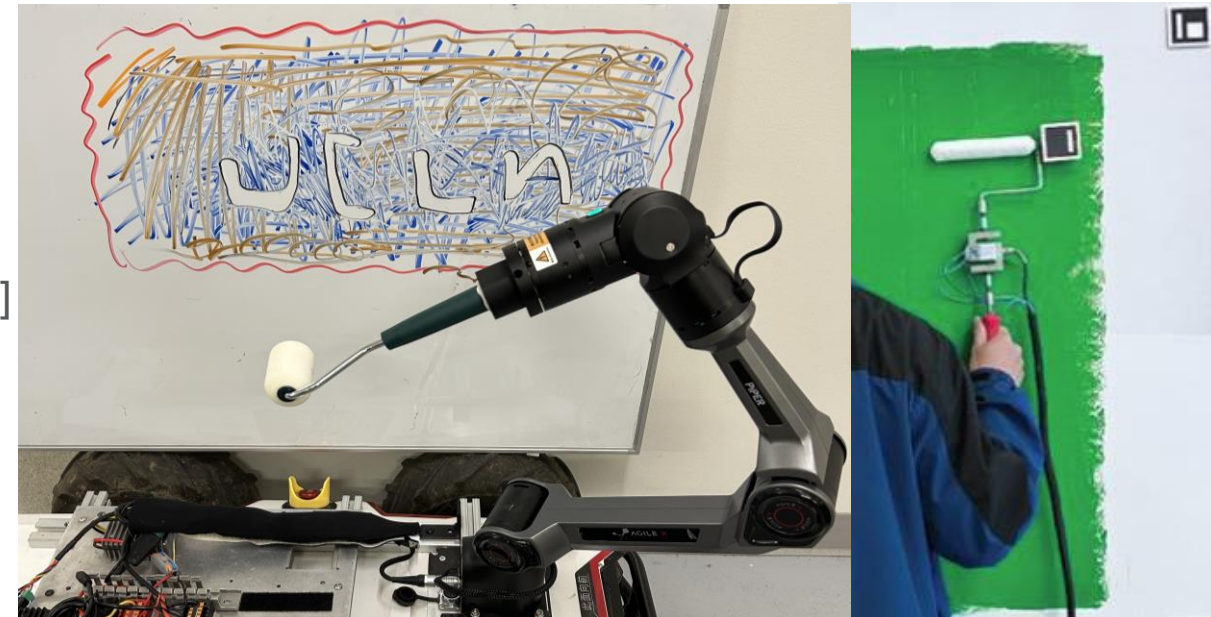
Stable brush–wall contact for robot painting

MOTIVATION:

- Contact-rich manipulation often requires force control [1]
- Contact force directly affects painting quality and safety. [2]
- Adaptive control is needed to handle surface variations [3]

OBJECTIVES OF CONTROLLER DESIGN :

Maintain stable brush–wall contact through real-time adaptive control



[1] M. Suomalainen, Y. Karayiannidis, and V. Kyrki, “A survey of robot manipulation in contact,” *Robotics and Autonomous Systems*, vol. 156, Art. no. 104224, 2022, doi: 10.1016/j.robot.2022.104224.

[2] K. Bansal, Y. Du, E. Palan, M. Quadir, and M. K. Jawed, “Robotic painting: mimicking human applicators,” *Journal of Coatings Technology and Research*, vol. 20, pp. 1369–1381, 2023, doi: 10.1007/s11998-022-00750-7.

[3] F. J. Abu-Dakka and M. Saveriano, “Variable impedance control and learning—A review,” *Frontiers in Robotics and AI*, vol. 7, Art. no. 590681, 2020, doi: 10.3389/frobt.2020.590681.



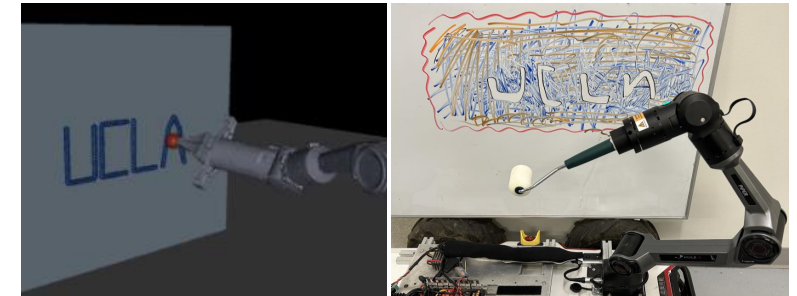
Methods and Control Architecture

SIMULATION:

$$M\ddot{e} + B\dot{e} + Ke = F_{\text{des}} - F_{\text{meas}}$$

$$e = x_{\text{imp}} - x_{\text{ref}} \quad e_F = F_{\text{des}} - F_{\text{meas}}$$

$$F_{\text{des}} - F_{\text{meas}} \rightarrow x_{\text{imp}} \rightarrow x_{\text{cmd}} \rightarrow \mathbf{x}_d \rightarrow \text{IK} \rightarrow q_{\text{cmd}}$$



REAL WORLD: JOINT-LEVEL IMPEDANCE CONTROL

$$\tau_i \approx Kp_i(q_{\text{ref}i} - q_i) + Kd_i(-\dot{q}_i).$$

MIT-based

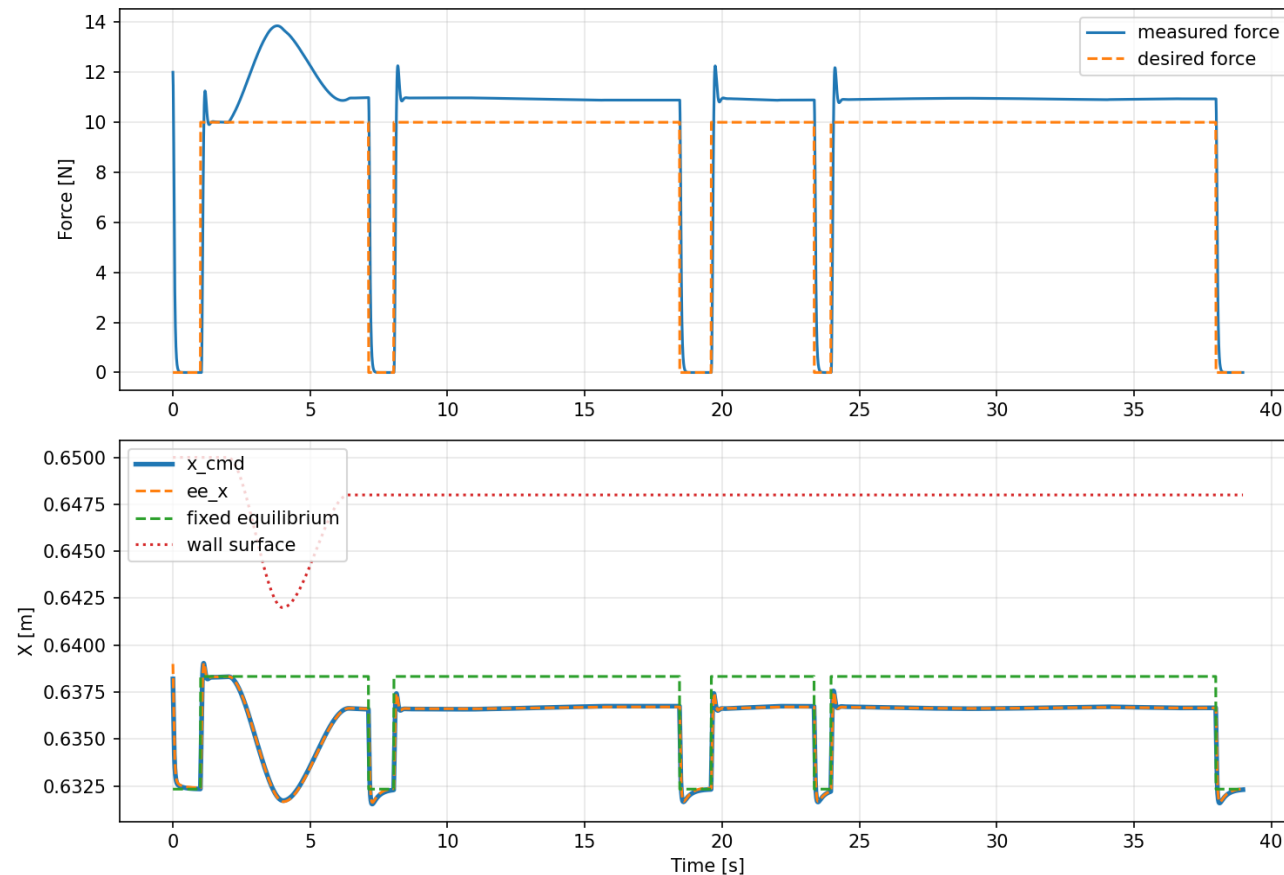
Each joint behaves as a virtual spring–damper.



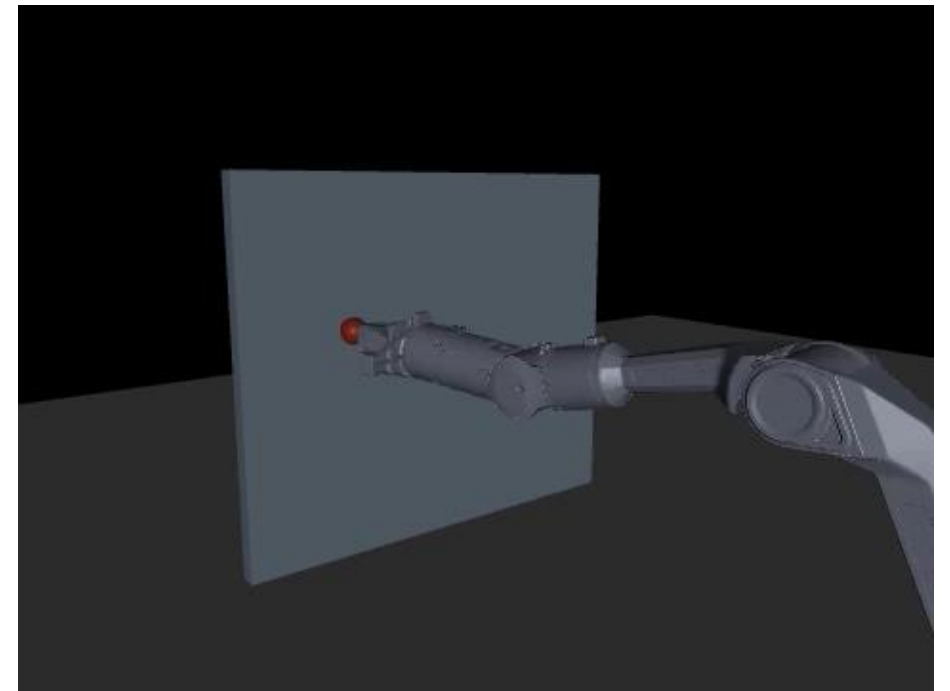
Results



SIMULATION:



$$M = 1.6 \text{ kg}, \quad B = 145 \text{ N} \cdot \text{s/m}, \quad K = 560 \text{ N/m}$$
$$F_{\text{des}} = 10 \text{ N}$$



Results

REAL WORLD:

- Control rate: 200 Hz
- Ramp time: 2 s
- Joint error limit: 0.35 rad $\approx 20^\circ$
- Letter speed: 0.02 m/s
- Stiffness scale: 0.3

$$K_p = [10, 8, 8, 8, 6, 6]$$

$$K_d = [0.8, 0.6, 0.6, 0.6, 0.5, 0.5]$$

$$K_p^{eff} = 0.3K_p = [3.0, 2.4, 2.4, 2.4, 1.8, 1.8]$$

$$K_d^{eff} = \sqrt{0.3}K_d \approx [0.44, 0.33, 0.33, 0.33, 0.27, 0.27]$$



Discussions

GAIN EFFECT TABLE

Gain	Increase effect	Risk
M (sim)	smoother command	slower response
B (sim)	less oscillation	more lag
K (sim)	less position drift	less compliance
Stiffness (real world)	Better tracking	Risk of damage

LIMITATIONS & CHALLENGES

Virtual path must stay behind the wall → limited workspace

Friction strongly affects low-stiffness impedance

High stiffness improves tracking but may damage the end effector



Future Directions

1. Track targets behind the wall under a maximum contact-force limit
2. Validate safety, tracking accuracy, and painting quality on real surfaces

